Shopping Mall Customer Segmentation and Spending Analysis

Business Insights Report

Tools Used: Excel, Power BI
Dataset: Shopping Mall Customer Segmentation Data
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Executive Summary

This project analyzes customer demographic and behavioral data to identify actionable customer segments for a shopping mall business. The analysis was performed using Power BI, with a focus on customer age, gender, annual income, and spending score.

The objective was to move beyond basic demographic reporting and identify practical business opportunities for customer targeting, retention, and marketing optimization. Customers were segmented into four income-spending groups to evaluate both current customer value and future growth potential.

The analysis found that gender-based spending differences are minimal, while income-spending segmentation provides stronger business insight. The most valuable customer group is the **High Income – High Spending** segment, which should be prioritized for premium retention strategies. The strongest conversion opportunity lies in the **High Income – Low Spending** segment, where customers have strong purchasing capacity but lower current engagement.

Overall, the analysis recommends a segment-based marketing approach to improve customer targeting, strengthen retention, and support better sales performance.

Data Preparation

The dataset was imported into Power BI and reviewed for quality and consistency before analysis. Key columns were validated, including Customer ID, Gender, Age, Annual Income, and Spending Score. Data types were checked and corrected where required.

Missing, null, and empty values were reviewed using Power Query filters and column profiling. Customer ID was treated as the unique customer identifier for customer-level analysis. Numerical fields such as Age, Annual Income, and Spending Score were reviewed to ensure they were suitable for analysis and segmentation.

After cleaning and validation, the dataset was used to build interactive dashboards, calculate key metrics, create customer segments, and generate business recommendations.

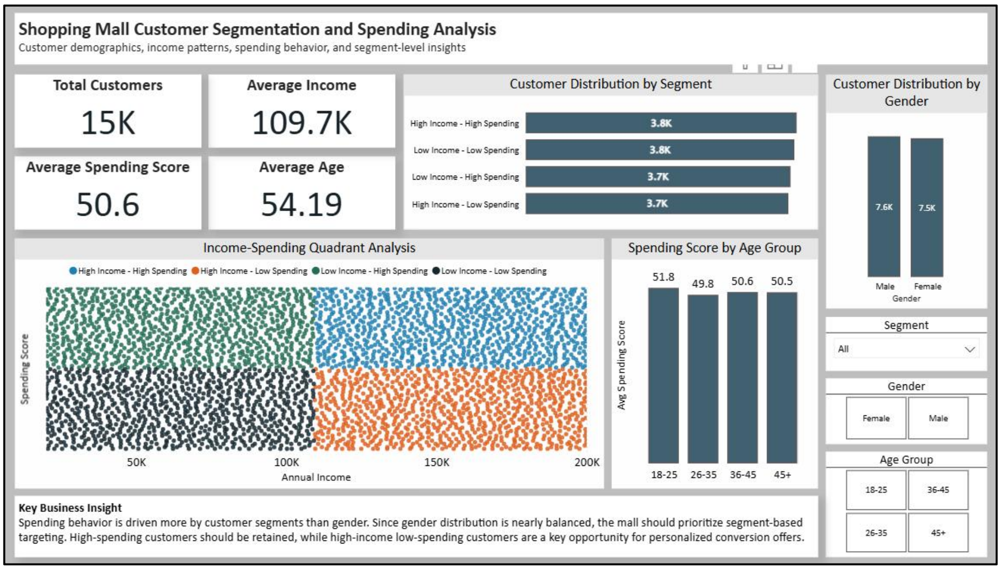
Dashboard Overview

Two Power BI dashboard pages were developed to support the analysis and business recommendations.

Dashboard Page 1: Executive Overview

This dashboard provides a high-level view of the customer base, including total customers, average age, average income, average spending score, gender-wise customer distribution, age-wise spending behavior, income-spending relationship, and segment-wise distribution.

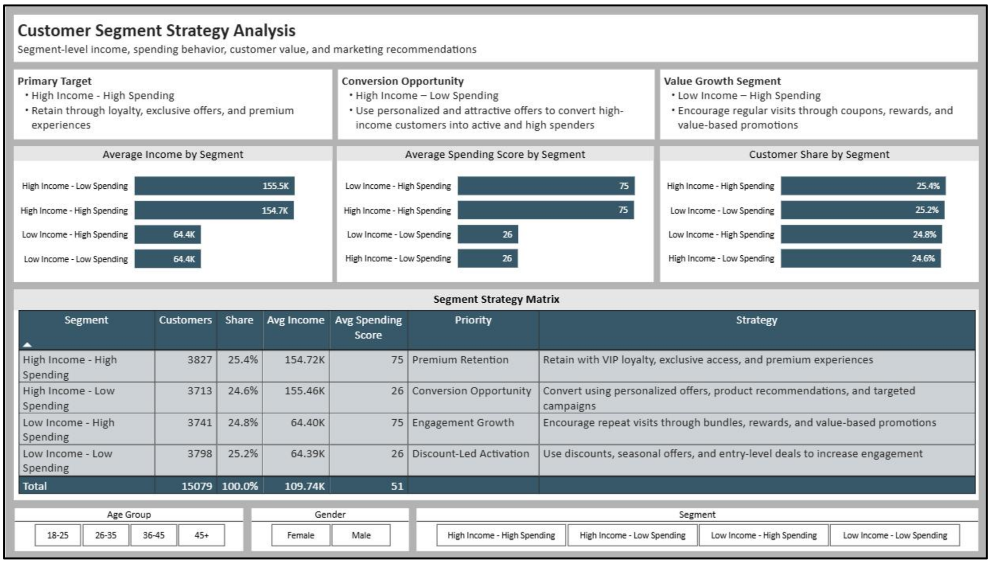
Figure 1: Executive Overview Dashboard



Dashboard Page 2: Segment Strategy

This dashboard focuses on segment-level performance and business action planning. It includes average income by segment, average spending score by segment, customer share by segment, segment priority, and recommended strategies for each customer group.

Figure 2: Segment Strategy Dashboard



Customer Behavior Analysis

Overall Customer Profile

The analysis includes **15,079 customers**. The average customer age is **54.19 years**, the average annual income is **109.74K**, and the average spending score is **50.59**.

These values indicate that the mall has a sizeable customer base with moderate to high purchasing capacity and balanced overall spending behavior. However, the average values alone do not fully explain customer behavior, which makes segmentation important for deeper analysis.

Age-Based Insight

The **18–25 age group** has the highest average spending score at **51.84**. This suggests that younger customers are slightly more active in terms of spending behavior. Campaigns focused on lifestyle, digital engagement, limited-time offers, and social shopping experiences may work well for this group.

At the same time, the overall average customer age is **54.19 years**, which indicates that the mall also serves a mature customer base. Therefore, marketing campaigns should not focus only on younger shoppers. The business should also maintain offers for mature customers, such as comfort-focused experiences, loyalty benefits, personalized recommendations, and premium service-based engagement.

Gender-Based Insight

Male customers have an average spending score of **50.8**, while female customers have an average spending score of **50.4**. The difference is very small, which suggests that gender does not create a meaningful difference in spending behavior.

From a business perspective, gender should be used as a supporting filter rather than the main basis for campaign planning. Customer segment, income level, spending score, and age group provide stronger insight for marketing decisions.

Income-Spending Relationship

The correlation between annual income and spending score is **0.00**, which indicates no meaningful linear relationship between the two variables.

This is an important business finding. It shows that high-income customers are not automatically high spenders. Some high-income customers show strong spending behavior, while others spend less. Because of this, customer segmentation is necessary to separate premium customers from conversion opportunities.

Segmentation Analysis

Customers were grouped into four income-spending segments:

Segment	Customers	Share	Avg Income	Avg Spending Score
High Income – High Spending	3,827	25.4%	154.72K	75.15
High Income – Low Spending	3,713	24.6%	155.46K	25.61
Low Income – High Spending	3,741	24.8%	64.40K	75.47
Low Income – Low Spending	3,798	25.2%	64.39K	25.76

The segment sizes are closely balanced, with each segment contributing roughly one-fourth of the customer base. This means the mall has multiple meaningful customer groups rather than one dominant segment. As a result, a single marketing strategy would not be effective for all customers. Each segment requires a different approach based on income level, spending behavior, and business potential.

Segment Interpretation:

High Income – High Spending
This segment represents premium customers. These customers have both high purchasing power and strong spending behavior. They are important for retention, repeat purchases, premium sales opportunities, and long-term customer value.

High Income – Low Spending
This segment has the highest average income but low spending behavior. These customers are financially capable but not fully engaged. This group should be treated as a major conversion opportunity because the business can potentially increase revenue by improving their engagement.

Low Income – High Spending
This group has lower income but high spending activity. These customers are already engaged with the mall and can be encouraged through value-based offers, rewards, bundled deals, and repeat-visit campaigns.

Low Income – Low Spending
This group has lower income and low spending behavior. These customers are likely to be price-sensitive and may require discounts, entry-level deals, coupons, or seasonal campaigns to improve engagement.

Business Recommendations

1. Primary Target Segment

The business should prioritize **High Income – High Spending** customers for retention and premium engagement. These customers already demonstrate high spending behavior and strong purchasing capacity, making them valuable for loyalty programs, premium services, exclusive offers, and VIP experiences.

Recommended actions:

- VIP loyalty programs
- Exclusive brand events
- Early access to sales or new collections
- Personalized premium offers
- Experience-based engagement campaigns

2. Growth Opportunity Segment

The **High Income – Low Spending** segment should be treated as the main conversion opportunity. These customers have high income but low spending activity, which means there is potential to increase revenue if the mall improves engagement.

Recommended actions:

- Personalized product recommendations
- Targeted premium offers
- Exclusive limited-time campaigns
- Personalized communication
- Upsell and cross-sell campaigns

3. Low-Spending Customer Strategy

Low-spending customers should not be targeted with one generic discount strategy. The approach should depend on the segment.

For **High Income – Low Spending** customers, offers should focus on personalization, premium value, exclusivity, and relevant recommendations. These customers have purchasing capacity, so the goal should be to improve engagement rather than only offering discounts.

For **Low Income – Low Spending** customers, offers should focus on affordability, coupons, seasonal discounts, entry-level promotions, and budget-friendly deals.

4. Retention Strategy for High-Income Customers

High-income customers can be retained by making the shopping experience more personalized and exclusive. Instead of depending only on discounts, the mall should focus on service quality, premium experiences, loyalty recognition, and targeted communication.

Retention strategies should create a sense of value, exclusivity, and convenience for high-income customers.

Final Business Conclusion

The analysis confirms that customer segmentation is more useful than basic demographic targeting. Gender-based spending differences are minimal, and income alone does not explain customer spending behavior.

The mall should adopt a segment-based marketing strategy. Premium customers should be retained through loyalty and experience-based initiatives. High-income low-spending customers should be converted through personalized campaigns and premium offers. Low-income high-spending customers should be encouraged through value-based rewards, while low-income low-spending customers should be activated through discount-led promotions.

This strategy can improve marketing efficiency, customer retention, and overall sales performance by aligning campaigns with customer behavior rather than relying only on broad demographic assumptions.